

Ethereal AI: Infrared Spectra of Polycyclic Aromatic Hydrocarbons with Machine Learning DFT Scaling Factors

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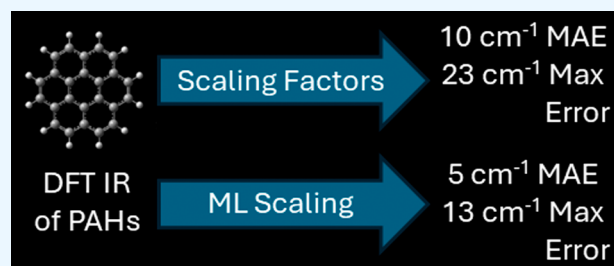


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ABSTRACT: Polycyclic aromatic hydrocarbons (PAHs) are of interest in astrochemistry and are studied through infrared (IR) signals from space. Deconvoluting the observed IR of PAHs often relies on computational chemistry for references. Current practice typically uses IR predicted by density functional theory adjusted with scaling factors to account for anharmonicity and basis set limitations. We present a machine learning approach to scaling factors that adjusts each predicted IR frequency individually based on computed frequencies, intensities, reduced masses, and force constants. Our machine learning model is able to obtain a mean absolute error (MAE) and maximum error of 5 and 13 cm^{-1} , respectively, a drastic improvement over a current-practice scaling factor approach with an MAE and maximum error of 10 and 23 cm^{-1} , respectively. Using machine learning to determine scaling of DFT-generated IR spectra shows promise for enabling higher quality predictions on larger systems than those that have previously been possible.



INTRODUCTION

Polycyclic aromatic hydrocarbons (PAHs) have been of significant interest in astrochemistry since they were determined to be responsible for many of the prominent infrared (IR) emission features observed in the IR spectra of interstellar space.^{1–3} PAHs are now believed to account for about 10% of the carbon in the universe,⁴ second only to carbon dioxide.⁵ Understanding whether PAHs play a role in the formation of other organic molecules,⁶ especially prebiotic compounds,⁷ is an ongoing discussion in the astrochemistry community. IR signatures from PAHs are also used in analyzing star-forming regions of space,^{8,9} the interstellar medium,¹⁰ and galactic evolution.^{11–13} PAHs have even been proposed as possible technosignatures, i.e., observable signs of extraterrestrial technology.¹⁴

Understanding PAHs in the universe requires interpreting IR data from telescopes and deconvoluting overlapping spectra from multiple chemical species, which motivated the development of the NASA Ames PAH Infrared Spectroscopic Database (PAHdb).^{15–17} The PAHdb includes 84 experimental IR spectra taken in argon matrices designed to imitate the low temperatures and pressures of the interstellar medium, but there are many additional PAH species that are not readily accessible on Earth.¹⁵ The PAHdb therefore also includes over 4000 theoretical IR spectra predicted with density functional theory (DFT).¹⁷ The PAHdb has been employed in many astrochemical studies, including to identify PAHs in a comet in our solar system's Oort cloud,¹⁸ in the atmosphere of Titan,¹⁹ and to interpret data from many other regions in space

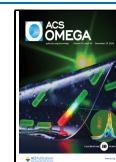
obtained with the Spitzer Space Telescope^{12,13} and the James Webb Space Telescope.^{9,11,20,21}

IR spectra are challenging to simulate with “spectroscopic accuracy” (errors below 1 cm^{-1}).⁵ A more complete accounting of the difficulties than we can address here can be found in a recent review by Fortenberry,⁵ but there are several challenges worth noting here: (1) Basis set incompleteness, in which errors arise from using truncations of a hypothetical infinite basis set, is typically addressed by extrapolation to the complete-basis-set (CBS) limit, but this is practical only for small systems. (2) Practical quantum chemical approaches require approximations to the full solution of the Schrodinger equation, which has been solved analytically only for the hydrogen atom. In wave function theory, the computational “gold standard” of the coupled cluster with singles, doubles, and perturbative triples [CCSD-(T)]²² is the least severe approximation in wide use, but it is computationally intensive and is not tractable for large systems. In the popular DFT,²³ all functionals in use can be viewed as approximations to the unknown “exact” functional. There are hundreds of approximate functionals available, and selection of the appropriate functional is a matter of ongoing discus-

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sion^{24–26} that is beyond the scope of this work. (3) More specific to vibrational spectroscopy, DFT and many wave function methods usually rely on the harmonic oscillator approximation for vibrational calculations, but real chemical bonds have an anharmonic potential with an increasingly shallow slope as the bond is stretched toward breaking.²⁷ For molecules with only a few atoms, it is possible to apply advanced methods like VPT2 that explicitly include anharmonicity²⁸ and can be extrapolated to the complete basis set limit, but that approach is not computationally feasible with any but the smallest of PAHs.⁵ For larger systems, such as those contained in the PAHdb, the common practice is to use DFT with a finite basis set, which systematically overestimates the IR frequencies.²⁷ The computed frequencies are then adjusted by multiplication by a “scaling factor”, which is typically a single value for a given combination of the DFT functional and the basis set determined by benchmarking known systems.²⁹ Scaling factors are primarily intended to correct for anharmonicity, but because they are derived for specific combinations of the method and basis set, they also mitigate the other errors noted here.²⁶ This is of particular importance in the context of PAHs because most astrochemical work⁵ and most methodological benchmarking are in consideration of much smaller systems, but with PAHs, the scale of the systems in question often precludes many best- or preferred-practice methods or basis sets. For example, the 2007 study by Merrick, Moran, and Radom²⁷ studied systems up to 10 atoms, and the recent VIBFREQ1295 database³⁰ contains systems up to 14 atoms. Conversely, naphthalene, the simplest PAH, contains 18 atoms, and there are many systems in PAHdb with hundreds of atoms. Therefore, the only feasible option is to derive scaling factors for a specific functional and basis set that can be applied to all systems under consideration, which is what Bauschlicher et al. have done.¹⁷

The most common practice with scaling factors is to find a single scalar value that minimizes disagreement between scaled computed frequencies for a large number of reference compounds,^{27,31} sometimes supplemented by high-quality theory such as some variant of CCSD(T).²⁶ The derived scaling factor can then be used by the community to correct calculations of IR spectra using the same computational method and the same basis set as was used in the original minimization. The well-known 1996 article by Scott and Radom provides a detailed mathematical description of the minimization.³¹ The most common approach has focused on minimizing root-mean-square error (RMSE),^{31,32} although Zapata Trujillo and McKemmish have called for using the median value instead.^{26,30} We present both within this work without making judgments about which statistical measures will be most relevant for the reader. Bauschlicher et al. used a variation on this scheme in the most recent version of the PAHdb because of their specific interest in reducing the maximum error, even at the expense of increasing small errors,¹⁷ and the maximum error is similarly highlighted in this work. The PAHdb adds a level of nuance to the traditional single scaling factor approach by applying three scaling factors rather than one,¹⁷ with the scaling factors being assigned to specific frequency ranges. Bauschlicher and Langhoff have been experimenting with multiple scaling factors for PAHs as early as 1997 in efforts to minimize error cations as well as neutral molecules,³³ both of which are present in the PAHdb, but more recently, multiple scaling factors have also been proposed by Zapata Trujillo and McKemmish in a general scheme for

the vibrational spectroscopy of other compounds.^{26,30} The three scale factors currently in use in the PAHdb were derived for several different basis sets for use with the B3LYP functional^{34–37} based on a gas-phase spectrum of naphthalene, the simplest and one of the best-understood PAHs.¹⁷ (The authors note that this using gas-phase data as their reference is a departure from their previous work using argon-matrix data but indicate the effects are limited.)¹⁷ With the 4-31G basis set,^{38–41} which is used for much of the PAHdb, the scaling factors are 0.956 for 0 to 1111.1 cm^{−1}, 0.952 for 1111.1 to 2500 cm^{−1}, and 0.960 for frequencies above 2500 cm^{−1}.¹⁷ This three-factor scheme has been noted to significantly improve accuracy for PAHs⁴ and has qualitative similarities to recommendations from more recent benchmarking work on smaller systems.²⁶ With their three-factor scheme, Bauschlicher et al. reported a mean average error (MAE) of 7.8 cm^{−1} and a maximum error of 26.9 cm^{−1} when using B3LYP/4-31G.¹⁷ They also reported using the BP86 functional^{42,43} and/or the 6-31G* basis set^{38,39,44,45} in cases where their preferred method of B3LYP/4-31G gave large errors.¹⁷

Bauschlicher et al.¹⁷ included several areas for future improvement, including:

- Calculation of more nitrogen-containing PAHs.
- Calculation of a wider variety of large PAHs.
- Calculations that better address anharmonicity.

As is often the case in computational chemistry, there is a tension between the need for more accurate calculations and the need to include larger systems. Recent work toward advancing the computational description of PAHs has generally addressed one side or the other of this tension: either proposing inexpensive methods to expand the scope to larger molecules^{46–48} or emphasizing anharmonicity more explicitly,^{49,50} which requires computationally expensive approaches.^{28,51} Attempting to follow modern recommendations^{26,30,32,52} to replace B3LYP with more advanced double-hybrid functionals or the 4-31G basis set with an augmented-polarized-split-valence basis set would increase computational cost without resolving the underlying tension. Such a strategy would work against the stated goal of PAHdb authors Bauschlicher et al.¹⁷ of including larger and more varied PAHs. Furthermore, changing the functional or basis set would entail a significant reimagining of PAHdb, which is beyond the scope of this work. Our hypothesis was that we could instead find a “middle way” and improve the accuracy of the relatively B3LYP/4-31G already in use by the PAHdb by using machine learning to derive case-specific scaling factors for each frequency individually.

Machine learning (sometimes referred to as “AI”) has been attempted in IR spectra as an alternative to performing any calculations,^{53,54} including for the IR spectra of PAHs.^{55–58} Machine learning has also been employed to predict new hypothetical species in astrochemistry.⁵⁹ In this work, we apply machine learning to obtain uniquely tailored scaling factors as corrections to DFT results. Rather than replacing the quantum chemical calculations outright, we propose using machine learning to make the existing scaling factor approach more effective. The traditional scaling factor has one input: the computed frequencies. The approach presented here instead uses several inputs: computed frequencies, intensities, force constants, and reduced masses. This work is distinct from the efforts of Kerkeni et al.,⁴⁶ in which a general “Frequency Scaling Function” was developed for semiempirical methods,

but in which DFT calculations were scaled using the same three scaling factors previously derived by Bauschlicher et al. for the PAHdb.¹⁷ Prior efforts by Bauschlicher et al. in ref 17 (and citations within) to improve accuracy included going from one scaling factor to two, and then three, with still more scaling factors being needed in the future noted as a strong possibility. Here, that concept is extended to its ultimate limit by employing machine learning to identify unique scaling factors for each computational band. To the best of our knowledge, this is the first use of machine learning for the computation of IR scaling factors.

Rather than minimizing error with one-three scalar parameters that are applied to large regions of the IR spectrum, as has been common practice, here we are proposing a system in which each calculated frequency is adjusted based on several input parameters that are readily available after a vibrational calculation. The goal of this study was to improve the quality of IR predictions for PAHs in the context of PAHdb with minimal disruption to the existing framework, but any questions of broader application or generalizability are beyond the scope of this work. With our individualized scaling factor approach, we hope to enable the study of larger PAHs with quantum mechanical methods, mitigate the need for untenably expensive anharmonicity-explicit methods, and simplify future work by reducing the need for multiple DFT functionals and basis sets on comparable systems.

RESULTS AND DISCUSSION

Table 1 shows the comparative test results of the different machine learning regression models. The best four, linear

Table 1. Comparative Performance of Machine Learning Regression Models on the Testing Portion of PAH IR Spectra Calculated with B3LYP/431-G. “RMSE” is Root Mean Squared Error and “MAE” is Mean Absolute Error. All Values are in Wavenumbers (cm^{-1})

model	RMSE	MAE	max error
SVR	6.10	5.07	13.17
elastic net	6.14	5.05	14.28
lasso	6.13	5.05	14.38
ridge	6.09	4.99	14.55
gradient boosting	8.31	5.82	35.73
MLP	19.44	13.64	66.52
k-NN	39.64	27.16	144.15
random forest	25.59	8.45	223.36
XGBoost	115.84	31.71	798.53

support vector regression (SVR), ElasticNet, Lasso, and Ridge, all perform similarly well. Elastic Net, Lasso, and Ridge are regularized versions of a linear combination of the features. Regularized models are constrained during the training process to avoid overfitting or “over learning” the training data, which leads to better generalization. The linear SVR combines the ideas of regularization and learning a linear combination of inputs with the added factor of using a fixed distance between the training samples and their predicted model values. Because SVR outperforms all the other models in the maximum error metric (the most important to minimize in this application) and is competitive in the other two metrics presented in Table 1, only SVR results are presented for the remainder of the discussion.

The feature coefficients for the SVR model are presented in Table 2. The computed frequency plays the largest role, which

Table 2. Feature Coefficients for the Linear SVR Model. These Coefficients are Multiplied by the Scaled Version of Each Feature, Which has Zero Mean and Unit Variance. The Absolute Values of These Coefficients Indicate Level of Influence in the Model’s Predictions

frequency	788.2629
relative intensity	−2.1628
intensity	0.8162
reduced mass	0.4069
force constant	0.1651

indicates the model still relies most heavily on the value that is the only input in traditional scaling factor approaches. Relative intensity is shown to be the second most important contributor. However, there are small contributions from the reduced mass and the force constant as well. This stands in contrast to what one might intuitively do when manually assigning peaks, namely, trying to align frequency and intensity with no other input variables. In keeping with other work on the vibrational frequencies of PAHs¹⁷ and general benchmarking,³² errors in intensities were not considered because predicting intensities is not of significant relevance to the application of interest. However, relative intensities are useful as an input value because they can favorably skew the model toward the most intense lines, which naturally are what will be most readily observed in astronomical data.

Figures 1 and 2 show the distribution of residuals (signed errors) in the SVR model’s test set and training set, respectively. In both cases, the C–H stretching region (i.e., the data points over 3000 cm^{-1}) is where the maximum error can be found, but those maximum errors are only slightly larger than the maximum in the region below 1650 cm^{-1} , and most errors in both regions are much smaller. The error in the C–H stretching region has previously been reported to be especially problematic in the study of PAHs,^{17,60} so bringing that error in closer alignment with the error in other regions of the spectrum is a noteworthy improvement.

Figure 3 shows the probability density distribution of the absolute error for the SVR model in both the training and test sets. Despite the presence of absolute errors of up to 18.19 cm^{-1} in the performance on the training set, their low probability is worth noting, which is below 0.025. As seen in Figure 2, only a few dots are above 15 or below −15. On the other hand, the model’s performance is most likely to present absolute errors below 13, with the highest probabilities for absolute errors being under 5. This confirms the reliability of the model to make experimental frequency predictions in all of the considered spectral regions. The difference in performance is also a product of using a regularized model, where generalization capabilities are prioritized over learning the training data.

Once we had a promising model, we applied it to the vast majority of the computational data in the PAHdb. This data is included in the Supporting Information. For the purposes of comparison with traditional scaling methods, the following discussion includes only the data points that were in the test set described above. This is to ensure an even comparison between traditional scaling and our model’s predictions and to

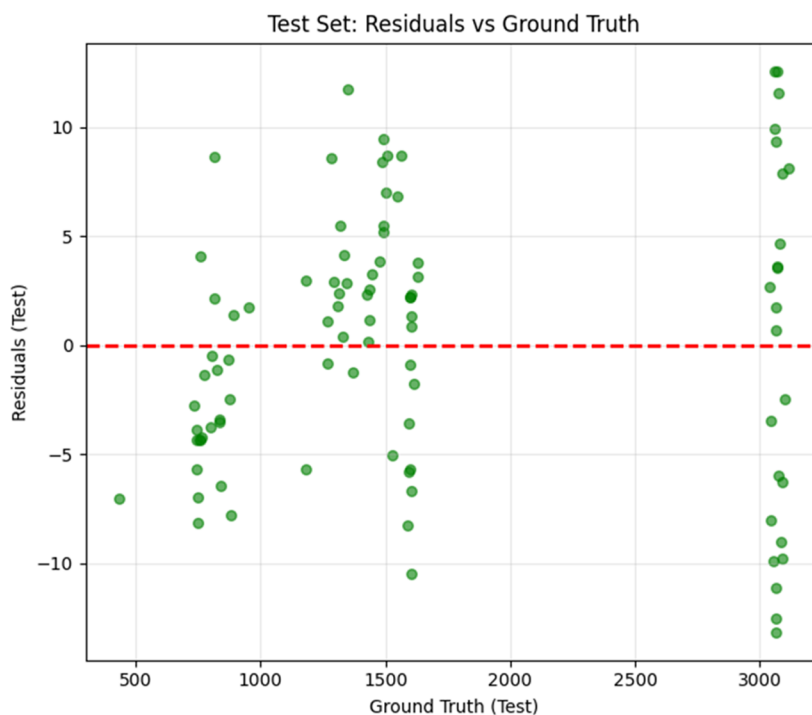


Figure 1. Residuals (signed errors) in the SVR model test set (20% of the total data points). The “ground truth” is the experimental wavenumber (in cm^{-1}).

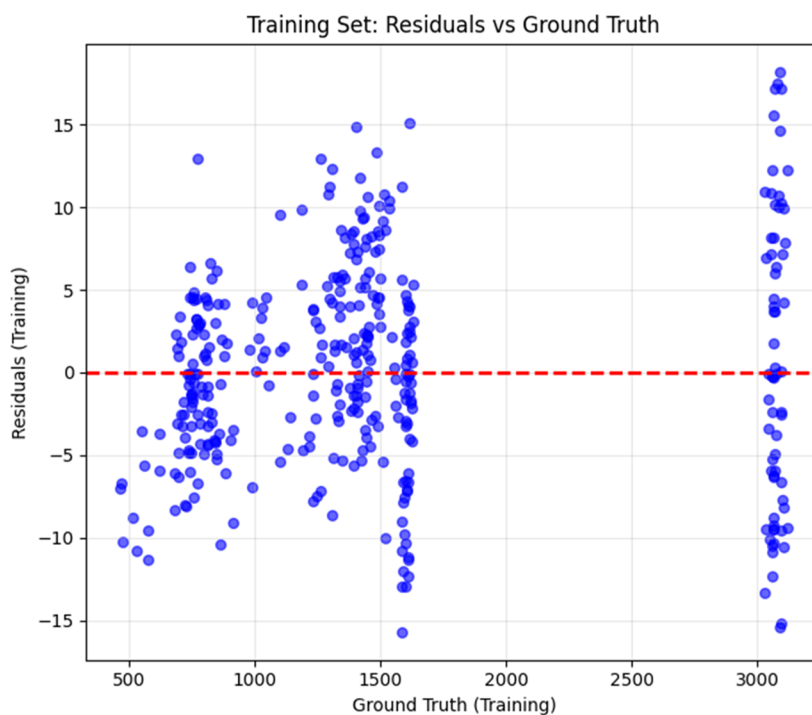


Figure 2. Residuals (signed errors) in the SVR model training set (80% of the total data points). The “ground truth” is the experimental wavenumber (in cm^{-1}).

enable more nuanced comparisons between subsets of the data.

A comparison between our model's results and traditional scaling is presented in Table 3. The “NIST scaling” was calculated based on the NIST Computational Chemistry Comparison and Benchmark DataBase (CCCBDB).⁶¹ The CCCBDB does not include a recommendation for B3LYP/4-

31G, so the 0.962 scaling factor for B3LYP/6-31G was used. The “PAHdb scaling” was calculated using the scaling factors provided for B3LYP/4-31G by Bauschlicher et al. in their article describing Version 3.00 of the PAHdb.¹⁷ Of particular note in Table 3 is that all error assessments are smallest for the SVR except for the median, which is only slightly smaller for the NIST scaling factor than for the SVR. Especially important

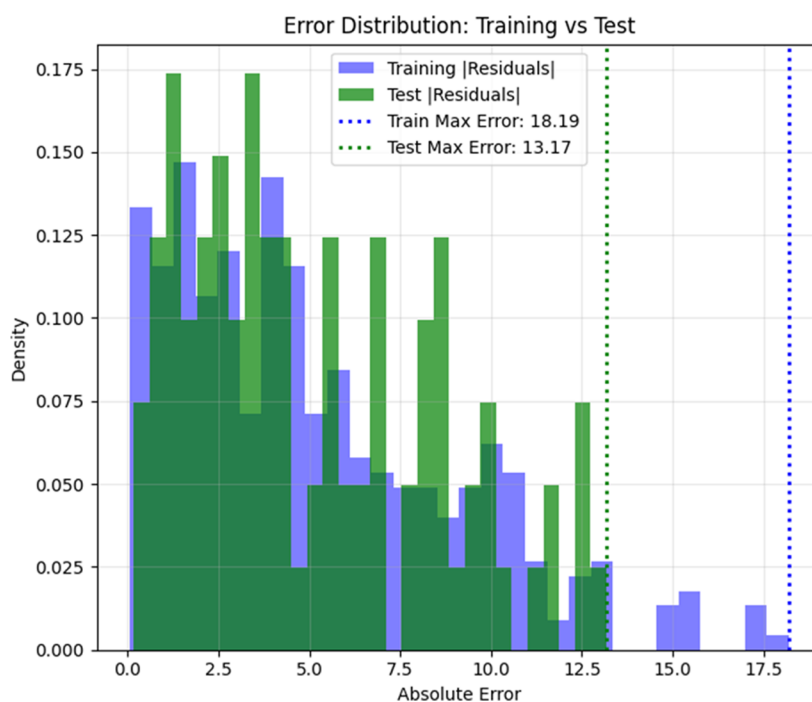


Figure 3. A comparison of the absolute errors (in cm^{-1}) between the training set (blue) and the test set (green).

Table 3. Comparison of Root Mean Squared Errors (RMSEs), Mean Absolute Errors (MAEs), Median Absolute Errors, and Maximum Errors of Traditional Scaling vs. Our Model All Values are in cm^{-1} Except for the Outliers, Which are Calculated as the Percentage of Entries with Absolute Errors over 10 cm^{-1}

	NIST scaling	PAHdb scaling	SVR
RMSE	9.56	12.41	6.10
MAE	6.40	10.41	5.07
median absolute error	3.81	10.75	4.13
outliers	18.3%	52.7%	8.6%
maximum error	29.92	23.49	13.17

is that the maximum error for the SVR is approximately 44 and 56% of the maximum error obtained for the same data using NIST scaling and PAHdb scaling, respectively. Reducing the maximum error in this way can reduce uncertainty in assigning astronomically observed bands and has been a stated priority of past work on the PAHdb.¹⁷ We also note that the percentage of outliers, calculated as the percentage of entries with absolute errors above 10 cm^{-1} , is a particular strength of SVR, with less than half of the outliers of NIST and less than one-sixth of the outliers of the PAHdb scaling. This implies that the reduced maximum error of SVR reflects the distribution of error as a whole.

Table 4 breaks up the MAEs into the wavenumber ranges used by Bauschlicher et al. in the PAHdb.¹⁷ The data are arranged in these ranges based on the experimental bands rather than the scaled or unscaled computational frequencies. It is clear that the C–H stretching region (greater than 2500 cm^{-1}) is generally the region of the greatest error, which aligns with previous notes by Bauschlicher et al.¹⁷ (among others, e.g., ref 60) that the C–H stretching has the greatest anharmonicity. This error is greatly reduced when using the SVR model in comparison to the traditional scaling factor approaches with MAEs 52% of the NIST scaling and 73% of

Table 4. Comparison of the Mean Absolute Errors (MAEs) of Traditional Scaling vs. Our Model, Including Subsets of Wavenumber Ranges Based on the Different Scaling Factor Ranges Used in the PAHdb¹⁷ All Values are in cm^{-1}

	NIST scaling	PAHdb scaling	SVR
All data	6.40	10.41	5.07
0–1111.1 cm^{-1}	4.23	3.03	4.03
1111.1–2500 cm^{-1}	3.23	14.66	4.31
2500+ cm^{-1}	14.47	10.26	7.53

the scaling factor used by the PAHdb for C–H stretching. In the middle region of 1111.1 to 2500 cm^{-1} , the SVR's MAE is similar to the MAE found using the NIST scaling, but both are much smaller than the MAE from the PAHdb scaling factor scheme. The MAEs are similar for all three approaches in the region below 1111.1 cm^{-1} . Table 5 provides a similar analysis

Table 5. Comparison of the Median Absolute Errors of Traditional Scaling vs. Our Model, Including Subsets of Wavenumber Ranges Based on the Different Scaling Factor Ranges Used in the PAHdb¹⁷ All Values are in cm^{-1}

	NIST scaling	PAHdb scaling	SVR
All data	3.81	10.75	4.13
0–1111.1 cm^{-1}	3.76	1.97	3.88
1111.1–2500 cm^{-1}	2.06	15.31	3.43
2500+ cm^{-1}	15.16	8.72	8.08

of the median absolute error, which is not substantially different from Table 4, except some of the differences become slightly more emphatic. Table 6 breaks up the maximum errors according to the same wavenumber regions. The SVR model consistently has lower maximum errors than both the NIST and PAHdb scaling, often by approximately a factor of 2. We note that our maximum error of 13.17 cm^{-1} is also a roughly 2-fold improvement over a general limit of 25 cm^{-1} for

Table 6. Comparison of the Maximum Errors of Traditional Scaling vs. Our Model, Including Subsets of Wavenumber Ranges Based on the Different Scaling Factor Ranges Used in the PAHdb¹⁷ All Values are in cm^{-1}

	NIST scaling	PAHdb scaling	SVR
All data	29.92	23.49	13.17
0–1111.1 cm^{-1}	9.14	11.63	8.65
1111.1–2500 cm^{-1}	12.15	22.60	11.75
2500+ cm^{-1}	29.92	23.49	13.17

traditional scaling factors determined by Zapata Trujillo and McKemmish.³²

It is worth noting that the primary limitation of this study was the availability of experimental reference data under the appropriate conditions. These data are biased toward small PAHs, and therefore, our confidence is stronger for small PAHs than for larger systems. The experimental data are also dominated by planar systems, although there were a small number of slightly nonplanar systems, as well. Additional detail is in the [Supporting Information](#). Should significant additional experimental work become available, it might be possible to retrain the SVR model in pursuit of increased confidence in the extension of our model to larger PAHs and PAHs of more diverse geometries.

CONCLUSION

The novel use of machine learning to determine individualized scaling for DFT-predicted IR is shown to be effective in reducing error for PAHs calculated with B3LYP/4-31G. The mean absolute error using the current PAHdb scaling factor scheme is 10.41 cm^{-1} , while the SVR model results presented here yield an MAE of 5.07 cm^{-1} . The maximum error, which is of particular importance, is reduced from 23.49 to 13.17 cm^{-1} in the current PAHdb scheme vs the SVR scheme, respectively. Comparisons within specific wavenumber ranges show that the machine learning prediction is especially advantageous for the C–H stretching region. This work could serve to enable consideration of larger compounds than those currently in the PAHdb by making B3LYP/4-31G feasible for a wider range of systems. Almost all of the 4000+ compounds in the theoretical section of PAHdb have now been recalculated using this machine learning protocol, and these data are provided in the [Supporting Information](#).

The work presented here includes several innovations, including the application of machine learning to deriving scaling factors, the individualization of scaling factors, and the inclusion of reduced masses and force constants in the derivation of the scaling factors. In our highly specific application and context, we achieved a maximum error (13.17 cm^{-1}) well below a generalized limit of traditional single scaling factor approaches (25 cm^{-1}).³² We make no claims based on the present data as to whether our approach can be extended to systems beyond PAHs or to methodological approaches beyond B3LYP/4-31G. However, the generalizability of this model is an open question for future exploration. Future work includes exploring this approach with a more general set of chemical systems and considering the functional and basis set as input variables for a more robust machine learning model. We anticipate the recent work of Zapata Trujillo and McKemmish on model chemistries in harmonic frequency calculations to be a useful guidepost in such further development.²⁶ Work in this area would need to

be done to determine if machine learning could be successfully applied to scaling factors for DFT predictions of IR spectra in a wide range of chemical contexts beyond PAHs, or even beyond astrochemistry.

METHODS

Experimental bands were obtained from the NASA Ames Research Center PAH IR Spectral Database (PAHdb).^{15,17,62–73} More details about the processing and specific identities of the PAHs are provided in the [Supporting Information](#). All DFT calculations were performed in Gaussian 16.⁷⁴ GaussView⁷⁵ was used for visualizations. In keeping with the vast majority of the work in the PAHdb,¹⁷ the B3LYP functional^{34–37} and the 4-31G basis set^{38–41} were used. Computational work began with B3LYP/4-31G geometry optimizations and frequency calculations using initial geometries from the computational entries in the PAHdb.¹⁷ Experimental and computational bands were matched automatically using a custom Python script up to 1590 cm^{-1} , above which bands were matched manually. Further details are provided in the [Supporting Information](#).

Several machine learning regression models were trained by using 465 correlated instances of computed features and their corresponding experimental bands from over 80 PAHs. The set of computed features comprises the frequency, intensity, relative intensity, reduced mass, and force constant. We divided the data set by setting 80% (372 instances) for training the models and the remaining 20% (93 instances) to test them. Each of these partitions comes from random sampling all portions of the spectrum and from all molecules in our data set. Before training our models, we standardized the features in both the training and testing partitions. This standardization process guarantees that each feature has zero mean and unit variance, reducing the effects of model bias toward features with larger values.

All machine learning work was conducted with NumPy,⁷⁶ Pandas,⁷⁷ and scikit-learn⁷⁸ using Google Colab.⁷⁹ ChatGPT⁸⁰ and Google Gemini⁸¹ were used to assist coding, but no AI tool was used in the writing of this manuscript. A variety of machine learning models were trained and tested, including SVR,⁸² Elastic Net regression,⁸³ Lasso regression,⁸⁴ Ridge regression,⁸⁵ Gradient Boosting,⁸⁶ MLP regression,⁸⁷ k-NN,⁸⁸ Random Forest regression,⁸⁹ and XGBoost,⁹⁰ all of which are implemented within scikit-learn.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c10225>.

PAHdb-SVR (XLSX)

Training (XLSX)

Machine (ZIP)

Scripts (ZIP)

Inputs (ZIP)

Coordinates (TXT)

Additional methodological detail, including the specific PAHs studied. All Gaussian inputs and outputs; the Jupyter notebook with our data analysis and preprocessing, as well as model training, selection, and performance analysis; data files used to train or process data with machine learning models; Pickle files to enable use of machine learning models without repeating the training,

including the feature standardizer; Python scripts used for data processing; and machine learning-predicted frequencies for most of the PAHdb systems, together with unscaled computed frequencies, intensities, force constants, and reduced masses, and also electronic and free energies for each system (PDF)

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Notes

The authors declare no competing financial interest.

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